NARAYANA IIT ACADEMY INDIA

Sec: Sr. IIT **Date: 25-02-24**

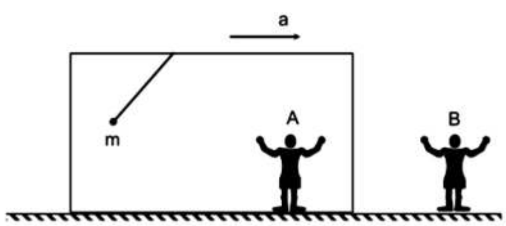
Daily Practices Sheet - 05

PHYSICS

1. A photon of wavelength λ is incident on a photoelectric plate and the most energetic electron from photoelectric plate is incident on a target metal of X-ray tube. If the minimum wavelength of photon emitted from target material is 2λ then work function of photoelectric plate is :-

1) $\frac{hc}{\lambda}$ 2) $\frac{hc}{2\lambda}$ 3) $\frac{hc}{4\lambda}$ 4) $\frac{hc}{8\lambda}$

2. A pendulum of mass 'm' is connected to the ceiling of a car which in turn moves with constant acceleration $\vec{a}$ on horizontal ground. Pendulum is at rest with respect to the car. There are two observers, A sitting in the car and B is on the ground at rest. Based on above information choose correct option. ($\vec{T}$ is tension force by on mass by the string)



1) Net force on mass as seen by A is $m\vec{g}+\vec{T}+m\vec{a}$

2) Net force on mass as seen by B is $m\vec{g}+\vec{T}+m\vec{a}$

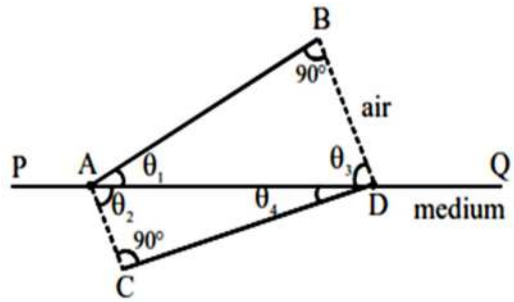
3) Net force on mass as seen by B is $m\vec{g}+\vec{T}$

4) Net force on mass as seen by A is $m\vec{g}+\vec{T}$

3. At $20°C$, a pipe open at both ends resonates at a frequency of 400 hertz. At what frequency does the same pipe resonates on a particularly cold day when the speed of sound is 3 percent lower than it would be at $20°C$?

1) 400 Hz 2) 388 Hz 3) 380 Hz 4) 366 Hz

4. The shown figure represents a wave front AB which passes from air to another transparent medium and produces a new wave front CD after refraction. The refractive index of the medium is (PQ is the boundary between air and the medium).



1) $\frac{\cos\theta_1}{\cos\theta_4}$ 2) $\frac{\cos\theta_4}{\cos\theta_1}$ 3) $\frac{\sin\theta_1}{\sin\theta_4}$ 4) $\frac{\sin\theta_2}{\sin\theta_3}$

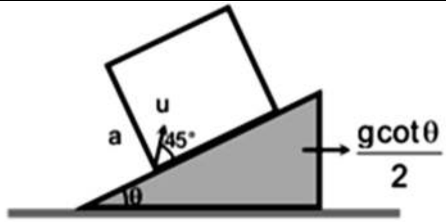
5. A cylindrical vessel of area of cross-section A is filled with water to a height H. It has a capillary tube of length l and radius r fitted horizontally at its bottom. If the coefficient of viscosity of water is η, then time required in which level of water will fall to a height $\frac{H}{2}$ is (density of water is ρ)

1) $\frac{\eta l r^2 4}{4A\rho} ln(2)$ 2) $\frac{4\eta l r^4}{\pi g A\rho} ln\left(\frac{1}{2}\right)$ 3) $\frac{8\eta l A}{\rho\pi g r^4} ln(2)$ 4) $\frac{4H\eta l\rho}{\pi g r^4} ln(2)$

6. A spring is placed between the jaws of screw gauge such that the spring is not all compressed. The main scale reads 2 divisions and circular scale reads 28 divisions. Now we turn the circular scale by $18°$ such that the spring is compressed. The circular scale has 200 divisions and the least count of the main scale is 1mm. What is the force exerted by the spring on the jaws if the spring constant is 100 N/m.

1) 4 mN 2) 3 mN 3) 5 mN 4) 7 Mn

7. The smooth wedge is accelerated at $\frac{g\cot\theta}{2}$ to the right. A cubical box of side a is moving on it. Inside the box a particle is projected with speed u relative to the box at an angle of 45^0 as shown. Find the time after which the particle will hit the box. (Assuming a is large, so that particle does not collide with top)

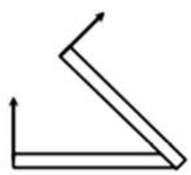


1) $\dfrac{u}{\sqrt{2}g\sin\theta}$ 2) $\dfrac{2\sqrt{2}u}{g\cos\theta}$ 3) $\dfrac{u}{g\sin\theta}$ 4) $\dfrac{u}{\sqrt{2}g\cos\theta}$

8. An isolated and charged spherical soap bubble has a radius ‘r’ and the pressure inside is 1 atmosphere. If ‘T’ is the surface tension of soap solution, then charge on the soap bubble is :-

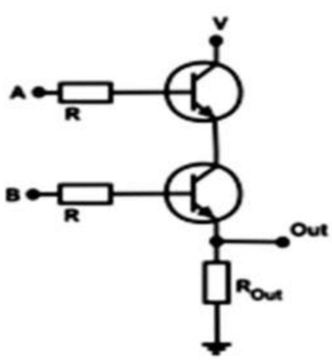
1) $2\sqrt{\dfrac{2rT}{\epsilon_0}}$ 2) $8\pi r\sqrt{2rT\epsilon_0}$ 3) $8\pi r\sqrt{rT\epsilon_0}$ 4) $8\pi r\sqrt{\dfrac{2rT}{\epsilon_0}}$

9. A long uniform rod lies flat on the table as shown. One end of the rod is slowly pulled up by a force that remains perpendicular to the rod at all times. The rod is to be brought to the vertical position without any slipping of the bottom end? For this to be possible, the coefficient of static friction between rod and ground may be?



1) 0.1 2) 0.25 3) 0.3 4) 0.5

10. The following transistor circuit is equivalent to which logic gate?

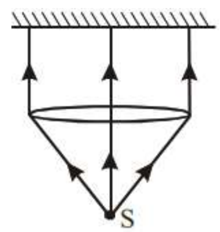


1) OR 2) NAND 3) XOR 4) AND

11. A particle of m = 2kg is moving with constant speed 15 m/s along a curved path $y = x^2$. Find its angular momentum about origin O when its velocity vector is at $53°$ with x-axis (in kg m^2/s)

1) 4 2) 8 3) 12 4) 16

12. A totally reflecting small plane mirror placed horizontally faces a parallel beam of light as shown in figure. The mass of mirror is 20 gm. Assume that there is no absorption in the lens and that 30% of the light emitted by the source goes through the lens. Find the power of the source needed to support the weight of the mirror (take $g = 10\ m/s^2$) :-



1) 80 MW 2) 100 MW 3) 20 MW 4) 25 MW

13. Under suitable conditions two α particles can combine to produce a proton and a nuclide of Li^7. If minimum kinetic energy required by an α-particle is 0.6 MeV for the reaction to proceed then range of nuclear interaction is of the order (Assume one of the α-particle stays at rest)

1) 10^{-11} m 2) 10^{-14} m 3) 10^{-16} m 4) 10^{-19} m

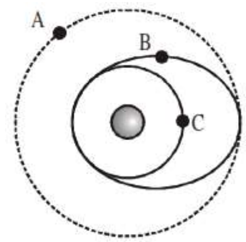
14. A conducting wire of length 'l' is placed on a rough horizontal surface, where a uniform horizontal magnetic field B perpendicular to the length of the wire exists. Least values of the horizontal forces required to move the rod when a current 'I' is established in the rod are observed to be $F_1 \& F_2 (< F_1)$ for the two possible directions of the current through the rod respectively. What is the coefficient of friction between the rod and the surface?

1) $\mu = \frac{F_1 - F_2}{BIl}$ 2) $\mu = \frac{F_1 + F_2}{BIl}$ 3) $\mu = \frac{F_1 + F_2}{2BIl}$ 4) $\mu = \frac{F_1 - F_2}{2BIl}$

15. A projectile is required to hit a target whose co-ordinates relative to horizontal and vertical axes through the point of projection are (α, β). If the bullet speed is $\sqrt{2g\alpha}$, then the condition when it is impossible to hit the target is

1) $3\alpha > 4\beta$ 2) $4\beta > 3\alpha$ 3) $3\beta > 4\alpha$ 4) $4\alpha > 3\beta$

16. Three equal mass satellites A, B, and C are in coplanar orbits around a planet as shown in the figure. The magnitudes of the angular momenta of the satellites as measured about the planet are L_A, L_B, and L_C. Which of the following statements is correct ?



1) $L_A > L_B > L_C$ 2) $L_C > L_B > L_A$ 3) $L_B > L_C > L_A$ 4) $L_B > L_A > L_C$

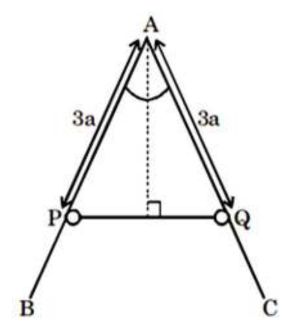
17. In optical bench experiment index error is +1 cm and -1 cm, between object needle and lens, lens and image needle respectively. Observed values of u and v are 9 cm and 17 cm. Focal length of lens is closest to:-

1) 6.15 cm 2) 5.54 cm 3) 5.88 cm 4) 6.25 cm

18. A telescope has an objective of focal length 100 cm and an eyepiece of focal length 2.5 cm. If you look into the objective (that is, into the wrong end) of this telescope, you will see distant objects reduced in size. By what factor will the angular size of objects be reduced?

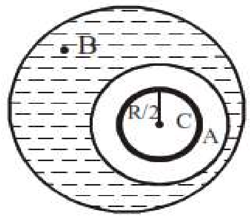
1) 40 2) 10 3) 20 4) 4

19. In the diagram, BAC is a rigid fixed rough wire frame in vertical plane, and angle BAC is $60°$. P and Q are two identical rings of mass m connected by a light elastic string of natural length 2a and elastic constant $\frac{mg}{a}$. If P and Q are in equilibrium when PA = AQ = 3a then the least coefficient of friction between the ring and the wire is μ. Then value of $\mu + \sqrt{3}$ is :-



1) 2 2) 3 3) 4 4) 7

20. Suppose there is an air bubble of radius $\frac{R}{2}$ inside the air cavity of radius R inside a water drop of radius 3R as shown in figure. The ratio of gauge pressure at point C to gauge pressure at point B is:



1) 4 2) 8 3) 12 4) 16

21. Which of the following is the most precise device for measuring length :-

1) a vernier calipers with 20 divisions on the sliding scale

2) a screw gauge of pitch 1 mm and 100 divisions on the circular scale

3) an optical scale that can measure length upto within a wavelength of visible light

4) a meter scale with divisions at each milimetre

22. Consider two identical circular loops of same radius, with the second (right side) coil rotated slightly clockwise relative to the first when looked from above as shown in figure (a). A large current is suddenly injected into the left side loop. What happen to the right side loop ?

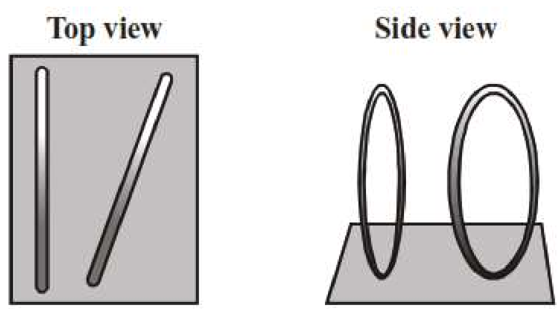
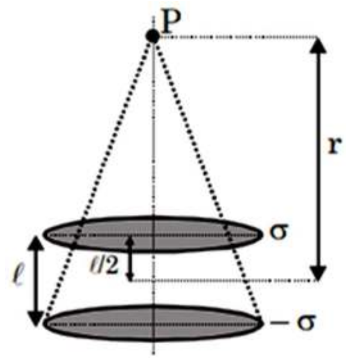


figure (a) figure (b)

1) Force to the left, torque rotates clockwise (in top view)
2) Force to the right, torque rotates clockwise (in top view)
3) Force to the right, torque rotates counterclockwise (in top view)
4) Can't tell without knowing which direction current injected into left loop

23. Two parallel disks each having radius R are separated by a distance l. The surface charge densities are σ and $-\sigma$. The electric field at point P, a large distance r along the axis of the disks is :



1) $\frac{\sigma R^2 l}{\varepsilon_0 r^3}$ 2) $\frac{2\sigma R^2 l}{\varepsilon_0 r^3}$ 3) $\frac{\sigma R^2 l}{4\varepsilon_0 r^3}$ 4) $\frac{\sigma R^2 l}{2\varepsilon_0 r^3}$

24. For amplitude modulated wave broadcast which of the following cases is most suitable

Wave (A) $A_m > A_C \quad \omega_m > \omega_C$

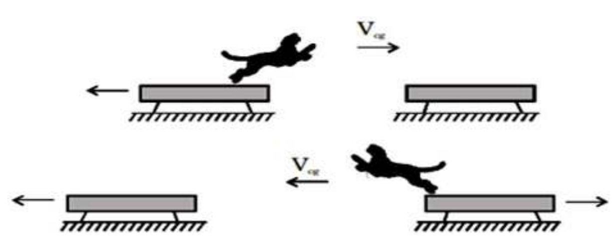
Wave (B) $A_m < A_C \quad \omega_m > \omega_C$

Wave (C) $A_m > A_C \quad \omega_m < \omega_C$

Wave (D) $A_m < A_C \quad \omega_m < \omega_C$

1) A 2) B 3) C 4) D

25. A strange cat with a mass m_C is sitting at rest on the left plank of a pair of identical planks. Each plank has mass m_S and they rest on frictionless ice. Suddenly, the cat leaps to the right plank, travelling with a horizontal speed v_{eg} measured with respect to the ground. The instant the cat reaches the right plank, it turns around and leaps back to the left plank. The horizontal component of the cat's speed is again v_{eg} measured with respect to the ground. The final speed of right plank in terms of the masses of the cat and planks and the cat's leaping speed is (The cat remains on the left plank after its return).

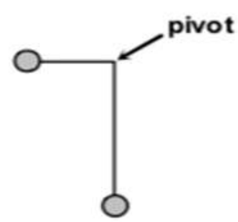


1) $\frac{2m_C v_{eg}}{m_S + m_C}$ 2) $\frac{2m_C v_{eg}}{m_S}$ 3) $\frac{m_C v_{eg}}{m_S + m_C}$ 4) $\frac{m_C v_{eg}}{m_S}$

26. Power transfer by a non-ideal cell to an external resistor can be same for two different values. Out of which one value (R_1) is less than internal resistance of cell and another (R_2) is greater than internal resistance of cell then :

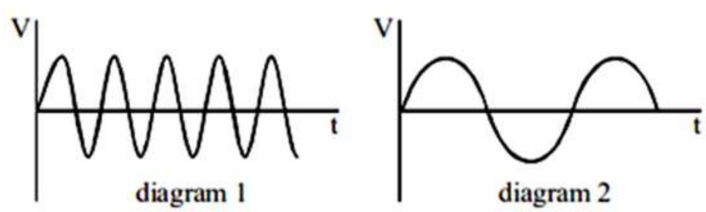
1) R_1 will give less internal loss
2) R_1 will give more battery life
3) R_2 will give less efficiency
4) R_2 will give more battery life

27. Two small spheres of mass m each are attached to the ends of a light rigid rod bent at a right angle and pivoted in the vertical plane at the vertex of the right angle. The vertical part of the rod is three times as long as the horizontal part. Initially, the rod is held at rest in the position shown in the diagram. The pivot is frictionless. Assume that masses of rods are negligible. Immediately after the system is released, which of the following options is incorrect?



1) Horizontal component of acceleration of center of mass of the system is 0.3 m/s^2 towards right
2) Horizontal component of hinge reaction is 0.3mg towards left
3) Vertical component of acceleration of center of mass of the system is 0.05 m/s^2 downwards
4) Vertical component of hinge reaction is 0.95mg upward

28. An series LCR circuit is resonating with a source whose emf varies with time as described in diagram-1. If we replace source by another source whose emf varies with time according to diagram-2, then :



1) for getting resonance again, decrease R
2) current will remain in phase with source voltage
3) for getting resonance again, decrease C
4) current will lead source voltage after replacing the source

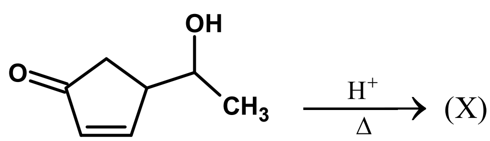
29. In the experiment of "finding resistance of Galvanometer by half deflection", we change the idea of taking the "half deflection" to "quarter deflection" keeping rest of the procedure same. The value of the resistance of high resistance box was set to R ohm and shunt resistance to obtain the quarter deflection was found to be S ohm. As usual, if $R >> S$, the value of galvanometer resistance determined is nearly.

1) $\frac{2S}{3}$ 2) S 3) $\frac{5S}{2}$ 4) 3S

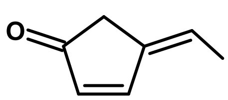
30. An unpolarized light passes through three polarizing sheets whose polarizing directions make an angle of $30°$, $60°$ and $30°$ with y axis in same sense. What fraction of initial intensity is transmitted by the system ?

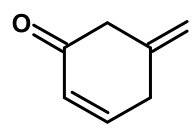
1) $\frac{1}{2}$ 2) $\frac{9}{32}$ 3) $\frac{3}{32}$ 4) $\frac{9}{64}$

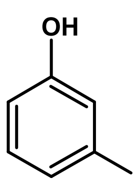
CHEMISTRY

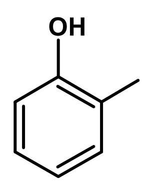
31. 

The major product (X) is

1) 

2) 

3) 

4) 

32. H_2O_2 solution used for hair bleaching is sold as solution of approx 5.0 gm H_2O_2 per 100 ml of the solution. The volume strength of the H_2O_2 is

1) 20 2) 16.69 3) 10 4) 27.5

33. The complexes $[Cr(NH_3)_6][Cr(SCN)_6]$ and $[Cr(NH_3)_4(SCN)_2][Cr(NH_3)_2(SCN)_4]$ are examples of

1) linkage isomers 2) ligand isomers

3) co-ordination isomers 4) ionisation isomers

34. Which of the following sequence is correct regarding the reactivity of acyl compounds towards nucleophilic attack in the order

1) acid chloride > amide > anhydride > ester

2) acid chloride > anhydride > ester > amide

3) ester > acid chloride > anhydride > amide

4) ester > anhydride > acid chloride > amide

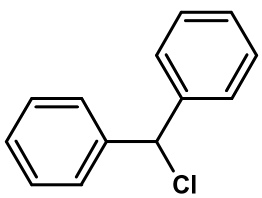
35. A solution contains Na_2CO_3 and $NaHCO_3$. 10 ml of the solution required 2.5 ml of 0.2 M HCl for neutralization using phenolphthalein as indicator. Methyl orange is then added when a further 2.5 ml of 0.4 M HCl was required. The amount of Na_2CO_3 and $NaHCO_3$ in 1 litre of the solution is

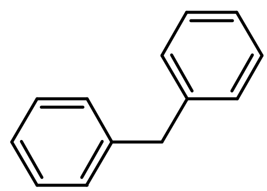
1) 5.3 g and 4.2 g
2) 3.3 g and 6.2 g
3) 4.2 g and 5.3 g
4) 6.2 g and 3.3 g

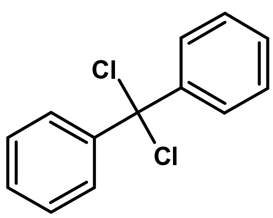
36. Mark out the incorrect match with the shape

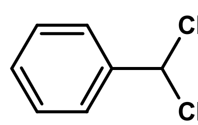
1) NH_2^-: Pyramidal
2) ICl_4^-: Square planar
3) $XeOF_2$: T-shape
4) XeF_2: linear

37. Which of the following is obtained as the product when excess benzene reacts with CH_2Cl_2 in the presence of anhydrous $AlCl_3$.

1) 

2) 

3) 

4) 

38. Consider the following sets of quantum numbers

	n	l	m	s
i	3	0	0	+1/2
ii	2	2	1	+1/2
iii	4	3	-2	-1/2
iv	1	0	-2	-1/2
v	3	2	3	+1/2

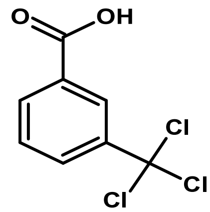
Which of the following sets of all electrons is not allowed set of quantum numbers?

1) I and iii
2) ii, iii and v
3) I, ii, iii and iv
4) ii, iv and v

39. Which of the following species exhibits the diamagnetic behavior?

1) NO 2) CO_2 3) O_2^- 4) O_2

40. The IUPAC name of



1) 3-Trichlorobenzoic acid 2) 3-(Trichloromethyl)-benzoic acid

3) 3-Chloralbenzoic acid 4) 3-Chlorobenzoic acid

41. Calculate the amount of $(NH_4)_2SO_4$ in 'g' which must be added to 500ml of 0.2 M NH_3 to yield a solution of pH=9.35. given $P^{K_b}(NH_3)=4.74$.(Given $10^{-0.09}=0.812$)

1) 8 g 2) 5.36 g 3) 4.55 g 4) 4.0 g

42. $(X)+C+Cl_2 \xrightarrow{1000K} (Y)+CO$

$(Y)+2H_2O \longrightarrow Be(OH)_2+2HCl$

Compound (Y) exist in polymeric chain structure and it is an electron deficient molecule. The compound (Y)

1) BeO 2) $BeO.Be(OH)_2$ 3) $Be(OH)_2$ 4) $BeCl_2$

43. Nucleoside is …………

1) a unit formed by attachment of Base to $3'$ position of sugar

2) a unit formed by attachment of Base to $1'$ position of sugar & phosphoric acid attached to $5'$ position of sugar

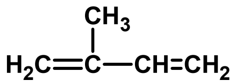
3) A unit formed by attachment of Base to $1'$ position of sugar

4) A unit formed by attachment of Base to $5'$position of sugar and phosphoric acid attached to $1'$ position of sugar

44. XY_2 dissociates as $XY_2(g) \rightleftharpoons XY(g)+Y(g)$. Initial pressure of XY_2 is 600 mm Hg The total pressure at equilibrium is 800 mm Hg. Assuming volume of system to remain constant the value of K_p in mm Hg is

1) 400 2) 200 3) 50 4) 100

45. Using bond energy data, calculate heat of formation of isoprene



Given bond energy of $C-H=98.8\,kcal$, $H-H=104\,kcal$, $C-C=83\,kcal, (C=C)=147\,kcal$ and ΔH of $C(s)\rightarrow C(g)=171\,kcal$

1) $28.3\,kcal\,mol^{-1}$ 2) $20.6\,kcal\,mol^{-1}$ 3) $283\,kcal\,mol^{-1}$ 4) $206\,kcal\,mol^{-1}$

46. Which of the following represents polyesters

(i) Dacron (ii) Terelene (iii) Orlon

1) I & iii 2) ii & iii 3) I & ii 4) iii only

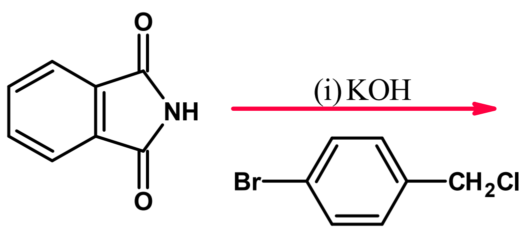
47. The de-broglie wavelength of an electron that has been accelerated through a potential difference of 100V is

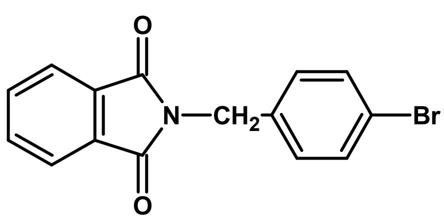
1) 0.1226 nm 2) 1.226 nm 3) 12.26 nm 4) 0.01226 nm

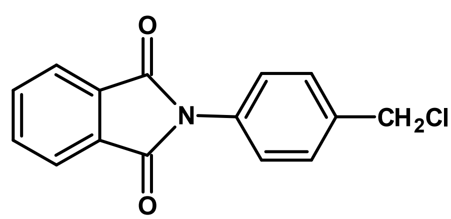
48. Gaseous cyclobutane isomerizes to butadiene in a first order process which has a k value at $153°C$ of $3.3\times10^{-4}s^{-1}$. How many minutes would it take for the isomerization to proceed 40% to completion at this temperature?

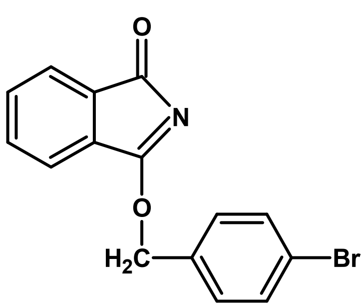
1) 26 min 2) 52 min 3) 13 min 4) 2.6 min

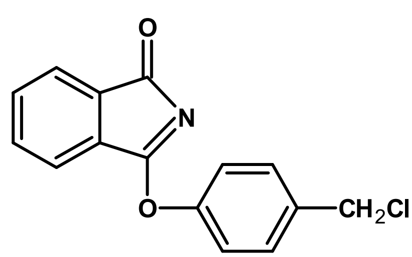
49. The major product of the following reaction is



1) 

2) 

3) 

4) 

50. Current efficiency is defined as the extent of a desired electrochemical reaction divided by the theoretical extent of the reaction times 100%. What is the current efficiency of an electrodeposition of Cu metal in which 9.8 g Cu is deposited by passage of 3A current for 10000 seconds. [Atomic weight of Cu = 63.5]

1) 0.9 % 2) 49.9 % 3) 99.1 % 4) 51.1%

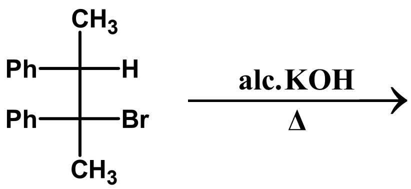
51. The K_p of a reaction is 10 atm at 300 K and 4 atm at 400K. The incorrect statement about the reaction following $A(g) \longrightarrow B(g) + C(g)$ is

1) The reaction is exothermic

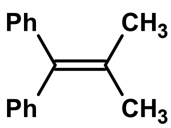
2) The E_a of forward reaction is more than that of backward reaction

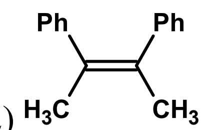
3) The rate of backward reaction increases more than that of forward reaction with increase of temperature

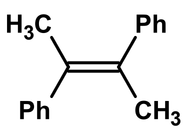
4) The difference between heat of reaction at constant pressure and that at constant volume is RT.

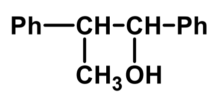
52. 

Major product of the reaction is:

1) 

2) 

3) 

4) 

53. $\frac{\Delta T_f}{K_f}$ has the same value for two different solutions i.e. 1 mole/kg. The first solution is 8% AB_2 by mass of solvent and second solution is 10% A_2B by mass of solvent. If both AB_2 and A_2B are non volatile and non electrolyte. Then Atomic masses of A and B will be respectively.

1) 20, 40 2) 20, 50 3) 40, 20 4) 50, 40

54. Xenon trioxide (XeO_3) forms xenate ion in alkaline medium. $XeO_3 + NaOH \rightarrow Na[HXeO_4]$. But the xenate ions slowly disproportionate in alkaline solution as $Na[HXeO_4] + NaOH \rightarrow Z + Xe + O_2 + H_2O$.

The compound Z is expected to be

1) Na_2XeO_3 2) Na_2XeO_4 3) Na_4XeO_6 4) Na_4XeO_4

55. Which of the following is not an artificial sweetner?

1) Aspartame 2) Sucrilose 3) Saccharin 4) Ranitidine

56. Find the incorrect statement:

1) Kraft temperature is the minimum temperature above which the formulation of micelles take place.

2) Soaps and detergents are associated colloids

3) Precipitating power of ions decrease in the order $Al^{3+} > Mg^{2+} > Na^{+}$

4) Gold sol are Macromolecular colloids formed by aggregation of many gold atoms

57. The correct order of matching of complex compound in column I with the properties in column II:

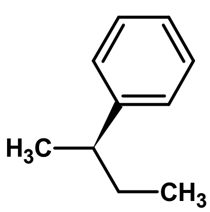
COLUMN I		COLUMN II	
A	$\left[Cr(NH_3)_6\right]^{3+}$	P	Tetrahedral and paramagnetic
B	$\left[Co(CN)_6\right]^{3-}$	Q	Octahedral and diamagnetic
C	$\left[Ni(CN)_4\right]^{2-}$	R	Octahedral and paramagnetic
D	$\left[Ni(Cl)_4\right]^{2-}$	S	Square planar and diamagnetic

1) A-R; B-Q; C-P; D-S
2) A-Q; B-R; C-P; D-S;
3) A-R; B-Q; C-S; D-P
4) A-Q; B-R; C-S; D-P

58. Which on photo chemical monochlorination gives only one enantiomeric pair besides other achiral products?

1) isopentane

2) Ethyl benzene

3) 

(Optically pure)

4) 2,3-di methyl pentane

59. Sodium oxide Na_2O is a crystalline solid. It does not have

1) Antiflourite structure

2) Na^{+} ions are present at body diagonal

3) Na^{+} ions are present at tetrahedral voids

4) O^{2-} ions are present at octahedral voids

60. Magnesium is mainly extracted by

1) Carbon reduction method

2) Self reduction method

3) The method of electrolysis in fused conditions

4) Leaching with aq. Solution of NaCN followed by reduction

MATHEMATICS

61. α,β,γ are the distinct roots of $x^3 = x^2 + 1$ the equation whose roots are $\alpha^2+\beta^3+\gamma^4, \beta^2+\gamma^3+\alpha^4, \gamma^2+\alpha^3+\beta^4$ is

1) $y^3-10y^2-33y-37=0$ 2) $y^3-10y^2+33y-37=0$

3) $y^3+10y^2+33y-37=0$ 4) $y^3+10y^2-33y-37=0$

62. There are 3 rows containing 2 seats in each row. The number of ways in which 3 persons can be seated such that no row remains empty is P then $\frac{P}{16}=$

1) 1 2) 2 3) 3 4) 4

63. Tangent are drawn to the circle $x^2+y^2=1$ at the points where it is meet by the circles $x^2+y^2-(\lambda+6)x+(8-2\lambda)y-3=0,$ λ-being parameter the locus of points of intersection of these tangents is $px+qy+10=0,$ Where $p+q=$

1) 1 2) 3 3) -1 4) -3

64. Let α^k where k=0,1,2,….,2013 are the 2014th roots of unity. If Z_1 and Z_2 be any two complex number such that $|Z_1|=|Z_2|=\frac{1}{\sqrt{2014}}$,then the value of $\sum_{k=0}^{2013}\left|Z_1+\alpha^k Z_2\right|^2$,is equal to

1) 4028 2) 0 3) 2 4) 2014

65. Let $I_1=\int_0^1\frac{dx}{1+\sqrt[3]{x}},\ I_2=\int_0^1\frac{dx}{1+\sqrt[4]{x}},$ then the value of $4I_1+3I_2$, is equal to

1) 3 2) 4 3)6 4) 7

66. A die is rolled 5 times. The probability there are at least two equal numbers among the outcomes obtained is

1) 319/324 2) 49/54 3) 13/18 4) 4/9

67. The minimum integral value of α for which the quadratic equation $\left(\cot^{-1}\alpha\right)x^2-\left(\tan^{-1}\alpha\right)^{\frac{3}{2}}x+2\left(\cot^{-1}\alpha\right)^2=0,$ has both positive roots,

1) 1 2) 2 3) 3 4) 4

68. If $\frac{1+x+x^2}{1-x}=a_0+a_1x+a_2x^2+a_3x^3+........$ and $\sum_{r=1}^{2010} a_r = p+q\times 10+r\times 100+s\times 1000$, where p,q,r,s are the digits of four digit number. $(p,q,r,s \in W)$ then

1) p = 6 2) S = 9 3) r = 0 4) q = 7

69. If $\cos 1+\cos 2+\cos 3+.........+\cos n = S_n$. That $\underset{n\to\infty}{\text{lt}} \frac{S_n}{n} =$

1) 1 2) 2

3) does not exists 4) 0

70. If $f(x)=f(x+1)+f(x+2)$ for all $x \in R$

$g(x)=\frac{x}{x-1}$ and

$g(g(x))=g^2(x),\ g(g(g(x)))=g^3(x),...........$

$f(f(x))=f^2(x),\ f(f(f(x)))=f^3(x),...........\&$

$f(1)=\frac{2}{3}$ and $f(g^{22}(3))=\frac{1}{3}$ then the value of $f(0)=$

1) $\frac{1}{3}$ 2) $\frac{2}{3}$ 3) 1 4) 0

71. A twice differentiable function $f(x)$ is defined for all real numbers and satisfies the following conditions $f(0)=2, f'(0)=-5$ and $f''(0)=3$ the function $g(x)$ is defined by $g(x)=e^{ax}+f(x)\ \forall x \in R$, Where a is any constant. If $g'(0)+g''(0)=0$ then sum of all possible values of a is

1) 1 2) -1 3) 2 4) 0

72. Points P,Q,R lie on same line. Three semi circles with the diameters PQ, QR, PR are drawn on same side of line segment PR. The centres of the semicircles are A,B,O respectively. A circle with centre C touches all 3 semi circles then the radius of this circle is (AQ=a, BQ=b)

1) $\frac{ab}{a+b}$ 2) $\frac{ab(a+b)}{a^2+b^2}$ 3) $\frac{ab(a+b)}{a^2+ab+b^2}$ 4) $\frac{ab(a+b)}{(a-b)^2}$

73. Statement:1 Contra positive of " If the weather is cold, then it will snow" is " If it will not snow, then the weather is not cold"

Statement: 2 Negation of " If it snows then they do not drive the car" is "It snows and they drive the car"

Which of the following is true, regarding the above statements?

1) Only S_1 2) Both S_1, S_2

3) Neither S_1 nor S_2 4) Only S_2

74. Find the smallest integer value of P for which the function $f(x) = 6Px - P\sin 4x - 5x - \sin 3x$ is monotonic increasing and has no critical point on R

1) 5 2) 4 3) 3 4) 6

75. If for defined real values of α, $\cot^3\alpha + \cot^2\alpha + \cot\alpha = 1$ then $\cos 2\alpha - \tan 2\alpha = ...$

1) 5 2) 1 3) 0 4) -1

76. If $\vec{x} = 3\hat{i} - 6\hat{j} - \hat{k}$, $\vec{y} = \hat{i} + 4\hat{j} - 3\hat{k}$, and $\vec{z} = 3\hat{i} - 4\hat{j} - 12\hat{k}$, then the magnitude of the scalar projection of $\vec{x} \times \vec{y}$ *on* $\vec{z}$ is:

1) 12 2) 15 3) 14 4) 13

77. Let $\pi_1 : x - y\sin\alpha - z\sin\beta = 0$,

$\pi_2 : x\sin\alpha - y + z\sin\gamma = 0$,

$\pi_3 : x\sin\beta + y\sin\gamma - z = 0$,

be 3 planes where $\sin\alpha, \sin\beta, \sin\gamma \neq 0$

S1: $\pi_1 \perp \pi_2$ if $2\sin\alpha = \sin\beta\sin\gamma$

S2:If π_1, π_2, π_3 are mutually perpendicular then $\sin\alpha.\sin\beta.\sin\gamma = 3$

S3: If $\alpha + \beta + \gamma = \frac{\pi}{2}$ then π_1, π_2, π_3 have a common line of intersection

S4: If $\alpha + \beta + \gamma = \frac{\pi}{2}$ then π_1, π_2, π_3 can never meet in a common line

then number of statements of the following that are correct

1) 1 2) 2 3) 4 4) 3

78. Let f(x) be a differentiable non-decreasing function such that

$$\int_0^x (f(t))^3\,dt = \frac{1}{x^2}\left(\int_0^x f(t)\,dt\right)^3 \forall x \in R-\{0\} \text{ and } f(1)=1.\text{If } \int_0^x f(t)\,dt = g(x)$$

then value of $\frac{xg'(x)}{g(x)}$

1) can be equal to 1
2) can be equal to 2
3) can be equal to -1
4) not independent of x

79. Let f(x) be a real valued function defined on the internal $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ such that

$$f(x) = e^{2x} + \int_0^{\sin x} \frac{e^t}{\cos^2 x + 2t\sin x - t^2}\,dt, \forall x \in \left(-\frac{\pi}{2},\frac{\pi}{2}\right).$$ Let g(x) be the inverse of f(x) where $0 \le x < \frac{\pi}{2}$ then the value of $f(0)+f'(0)+f''(0)$ equals

1) $g'(1)$
2) $3g'(1)$
3) $(g'(1))^{-2}$
4) $(g'(1))^{-1}$

80. If the system of linear equations $x+2ay+az=0, x+3by+bz=0, x+4cy+cz=0$ has a non-trivial solution & $a,b.c>0$ then the minimum value of $\left(\frac{a^3+c^3}{b^3}\right)$ is

1) 1
2) 0
3) 2
4) 3

81. If $S_n = 1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+............+\frac{1}{n}$ and $n>2$, then S_n would always be

1) more than $n(n+1)^{\frac{1}{n}}-n$
2) less than $n(n+1)^{\frac{1}{n}}-n$
3) equal to $n(n+1)^{\frac{1}{n}}-n$
4) None of these

82. $\bar{a},\bar{b},\bar{c}$ are non zero vectors such that $\bar{a}$ is perpendicular to both $\bar{b},\bar{c}$ and $|\bar{a}|=1,|\bar{b}|=2,|\bar{c}|=1,\bar{b}.\bar{c}=1$. There is a vector $\bar{r}$ coplanar with $\bar{a}+\bar{b}$ and $2\bar{b}-\bar{c}$ and $\bar{r}.\bar{a}=1$ then minimum value of $|\bar{r}|$ is

1) $2\sqrt{\frac{6}{5}}$
2) $\frac{\sqrt{5}}{2}$
3) $\frac{4}{\sqrt{13}}$
4) $\sqrt{\frac{6}{5}}$

83. If $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and $D = \begin{vmatrix} a_1+pb_1 & b_1+qc_1 & c_1+ra_1 \\ a_2+pb_2 & b_2+qc_2 & c_2+ra_2 \\ a_3+pb_3 & b_3+qc_3 & c_3+ra_3 \end{vmatrix}$ then

1) $D=\Delta$
2) $D=\Delta(1-pqr)$
3) $D=\Delta(1+pqr)$
4) $D=\Delta(1+p+q+r)$

84. For any three sets A, B and C, consider the following statements (where Δ denotes symmetric difference of the sets A and B)

A) $(A-B)\cup(B-A)=(A\cup B)-(A\cap B)$ B) $A\cap(B-C)=(A\cap B)-(A\cap C)$

C) $A\cap(B\Delta C)=(A\cap B)\Delta(A\cap C)$

Which of the following is correct

1) Options A and B only 2) Options B and C only

3) All the Options A,B and C 4)Options A and C only

85. Consider the function defined by $f(x)=e^{-x}$ on the interval $[0,10]$ let $n>1$ and let $0, x_1, x_2, x_3, ..., x_n$ be the numbers such that $0=x_1<x_2<x_3<.....<x_n=10$, which of the following is greatest

1) $\sum_{j=1}^{n} f(x_j)(x_j-x_{j-1})$ 2) $\sum_{j=1}^{n} f(x_{j-1})(x_j-x_{j-1})$

3) $\sum_{j=1}^{n} f\left(\frac{x_j+x_{j-1}}{2}\right)(x_j-x_{j-1})$ 4) $\int_0^{10} f(x)dx$

86. $\int e^x\left(\tan^{-1}x+\frac{2x}{(1+x^2)^2}\right)dx$ is equal to

1) $e^x\left(\tan^{-1}x-\frac{1}{1+x^2}\right)+C$ 2) $e^x\left(\tan^{-1}x+\frac{1}{1+x^2}\right)+C$

3) $e^x\left(\cot^{-1}x-\frac{1}{1+x^2}\right)+C$ 4) $e^x\left(\tan^{-1}x+\frac{2}{1+x^2}\right)+C$

87. If the equation of the chord joining the points $P(\theta)$ and $D\left(\theta+\frac{\pi}{2}\right)$ on $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is $x\cos\alpha+y\sin\alpha=p$ then $a^2\cos^2\alpha+b^2\sin^2\alpha=$

1) $4p^2$ 2) p^2 3) $\frac{p^2}{2}$ 4) $2p^2$

88. A circle with its centre at the focus of the parabola $y^2=4ax$ and touching its Directrix intersects the parabola at points A, B. Then length AB is equal to

1) 4a 2) 2a 3) a 4) 7a

89 . The locus of a point, from where tangents to the rectangular hyperbola $x^2-y^2=a^2$ contain an angle of 45^0, is

1) $(x^2+y^2)+a^2(x^2-y^2)=4a^2$

2) $2(x^2+y^2)+4a^2(x^2-y^2)=4a^2$

3) $(x^2+y^2)^2+4a^2(x^2-y^2)=4a^4$

4) $(x^2+y^2)^2+a^2(x^2-y^2)=a^4$

90. The mean of five observations is 4 and their variance is 5.2. If three of these observations are 1, 2 and 6, then the other two observations can be

1) 2 and 9 2) 3 and 8 3) 4 and 7 4) 5 and 6

KEY SHEET

1	**2**	2	**3**	3	**2**	4	**3**	5	**3**
6	**3**	7	**2**	8	**2**	9	**4**	10	**4**
11	**2**	12	**2**	13	**2**	14	**4**	15	**2**
16	**1**	17	**2**	18	**1**	19	**1**	20	**4**
21	**3**	22	**2**	23	**4**	24	**4**	25	**2**
26	**4**	27	**2**	28	**4**	29	**4**	30	**2**
31	**3**	32	**2**	33	**3**	34	**2**	35	**1**
36	**1**	37	**2**	38	**4**	39	**2**	40	**2**
41	**2**	42	**4**	43	**3**	44	**4**	45	**2**
46	**3**	47	**1**	48	**1**	49	**1**	50	**3**
51	**2**	52	**3**	53	**3**	54	**3**	55	**4**
56	**4**	57	**3**	58	**2**	59	**4**	60	**3**
61	**2**	62	**3**	63	**1**	64	**3**	65	**2**
66	**2**	67	**2**	68	**3**	69	**4**	70	**3**
71	**2**	72	**3**	73	**2**	74	**1**	75	**2**
76	**3**	77	**2**	78	**1**	79	**3**	80	**3**
81	**1**	82	**3**	83	**3**	84	**3**	85	**2**
86	**1**	87	**4**	88	**1**	89	**3**	90	**3**